Jiahao LI (Jarvis)

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https://github.com/JarvisLee0423

Education

City University of Hong Kong (CityUHK) **2024.09 - Present**
Ph.D. Student in Computer Science Supervisor: Prof. Jianping Wang
Research Direction: Autonomous Driving and 3D Scene Reconstruction

The Chinese University of Hong Kong (CUHK) **2021.09 - 2022.11**
Master of Science in Computer Science Supervisor: Prof. Kin Hong Wong
Research Direction: 2D Object Detection and Image Captioning

Hong Kong Baptist University (HKBU) **2017.09 - 2021.07**
Bachelor of Science in Computer Science and Technology Supervisor: Prof. Hui Zhang, Prof. Zhiyuan Li, Prof. Raymond Lee
Research Direction: Quantum Finance and Graph NN

Publications

[1] Li, J., Chen, X., Jiang, Z., Zhou, Q., Li, Y. H., & Wang, J. (2025). Global regulation and excitation via attention tuning for stereo matching. In Proceedings of the IEEE/CVF International Conference on Computer Vision (pp. 25539-25549).
[2] Huang, C., Liu, Y., **Li, J.**, Tian, H., Chen, H. (2023). Application of YOLOv5 for mask detection on IoT. Applied and Computational Engineering,29,1-11.
[3] **Lee, J.**, Huang, Z., Lin, L., Guo, Y., & Lee, R. (2023). Chaotic Bi-LSTM and attention HLCO predictor-based quantum price level fuzzy logic trading system. *Soft Computing*, *27*(18), 13405-13419.

Projects

The 17th F1Tenth Grand Prix (CPS-IoT 2024) **2024.04 - 2024.05**
FSM Speed Team Member Hong Kong
- Our team (FSM Speed) won the championship.
- In the lap time race, our team implemented a map-based algorithm involving an AMCL localization module, RaceLine generation module, and MPC control module.
- In the head-to-head competition, the classical follow the gap algorithm is optimized. The speed and steer-angle in the next timestamp are calibrated based on the farthest point from the ego vehicle and current speed.
- Project Page: https://xyunaaa.github.io/research/f1tenth/.

Adaptive Error Aware Cost Volume for Stereo Matching (AEACV-Stereo) **2023.10 - 2024.03**
First Author Beijing
- We propose a dynamic sampling strategy based on an error map to accelerate iterative stereo matching algorithm.
- We propose a noise-filtering cost volume method for improved disparity prediction accuracy in ill-posed regions.
- Code: https://github.com/JarvisLee0423/AEACV-Stereo.

Image Caption with Full-Transformer **2021.09 - 2022.06**
First Author Hong Kong
- Proposed an Image Captioning algorithm based on Full-Transformer.
- Proposed a BERT-based Encoder to process text information.
- Proposed an Encoder based on Vision-Outlooker to process image information.
- Code: https://github.com/JarvisLee0423/Captioning-and-Answering-with-Transformer.

Complex Networks and the Applications in Deep Learning **2020.09 - 2020.12**
First Author Zhuhai
- Apply complex graph theory (random graphs ER, WS, BA) to neural networks and construct random networks.
- Study the relationship between network computation graph topology and network performance.
- Network performance is measured and explored through time complexity and space complexity.
- Code: https://github.com/JarvisLee0423/RandWiredNN-Model.

Live Long and Prosper: How Herring and Mackerel Affect Scottish Fisheries **2020.03 - 2020.04**

Team Leader Zhuhai

- Mathematical Contest in Modeling (MCM 2020) Meritorious Winner.
- Use linear regression to predict sea surface temperature changes over the next 50 years to explore the impact of climate change on herring and mackerel migrations.
- Proposed an economic model of the relationship between fishing revenue and fishing location and protein decay using Gaussian distribution simulation.

Working Experience

ByteDance - PICO MR Team **2022.06 - 2024.02**

Computer Vision Algorithm Engineer (Full-Time) Beijing

- 3DoF PICO-VR Live Stream:
    - Implemented stereo disparity estimation system for distorted spherical cameras.
    - Implemented distortion direction exchange system to change the distortion direction from vertical to horizontal.
    - Fixed the OpenCV bug to generate depth from distorted spherical disparity and generate mesh from depth.
- Full 3D Object Reconstruction from two video views:
    - Applied consistency and robust depth estimation methods to generate depths for each frame of both videos.
    - Applied SLAM to generate pose for each frame of both videos.
    - Reconstructed half objects from each video.
    - Applied Point-Registration algorithm to match the half objects of each video to generate the full 3D object.
- Neural Network 3D Scene Reconstruction:
    - Applied Monocular Depth Network to estimate depth for each view of the scene.
    - Applied RAFT-based Stereo Matching Network to estimate disparity for each view of the scene.
- Neural Network based Depth Inference Accuracy upper boundary exploring:
    - Applied RAFT-based large vision model to explore the upper boundary of the stereo matching method.
    - Applied Multi-View Stereo based large vision model to explore the upper boundary of the depth estimation method.
    - Applied Monocular-based large vision model to explore the upper boundary of the depth estimation method.
    - Applied DINO and iBOT based large vision model to extract better vision feature for depth estimation tasks.
    - Used large model to distill the small model to get better performance.
- Depth Inference Code-base development and maintain:
    - Developed the depth inference codebase based on aipack, which is the code-engine in ByteDance.
    - Implemented useful backbone for vision tasks, like ResNet, EfficientNet, MobileNet, Vision Transformer and so on.
    - Implemented useful components for depth estimation tasks, like Cost Volume, Correlation, ConvGRU and so on.
    - Transferred all previous depth codebase with no accuracy dropping.

3D Print Lab **2020.06 - 2020.08**

Research Assistant (Intern) Zhuhai

- VR Exhibition Development:
    - Used Unity to implement a VR Exhibition system, including real-time viewing and editing.
    - Implemented the object edition function, including pick-up object, scale, rotation, translation, and wall absorption.
    - Implemented the view edition function, including view translation and projection, light-condition changing.